

Optimization Worksheet — Worked Answer Key

Sixteen substantial modeling problems with constraints, domains, critical points, and verification.

Revision: 2026-07-23 **Canonical page:** <https://bettergrades.net/subjects/math/calculus/worksheets/optimization/>

1. Squares of side x are cut from a 12 by 12 sheet. Maximize the open box volume.

Answer: $x = 2$, $V_{\max} = 128$

- $V = x(12-2x)^2$ to the power 2 on $0 < x < 6$; $V' = 12(x-2)(x-6)$, so the interior maximum is $x=2$.
- Check feasibility, endpoints, and the sign or second derivative before declaring the optimum.

2. Squares of side x are cut from a 20 by 12 sheet. Maximize volume.

Answer: $x = \frac{16-\sqrt{76}}{3} \approx 2.43$

- $V = x(20-2x)(12-2x)$. Solve $V' = 12x^2 - 128x + 240 = 0$ and retain the feasible critical point.
- Check feasibility, endpoints, and the sign or second derivative before declaring the optimum.

3. Use 240 m of fence for three sides of a riverside rectangle. Maximize area.

Answer: $x = 60$, $y = 120$, $A = 7200$

- With $2x+y=240$, $A = x(240-2x)$; $A' = 240-4x=0$.
- Check feasibility, endpoints, and the sign or second derivative before declaring the optimum.

4. A 600 m rectangle has one divider parallel to its width. Maximize area.

Answer: $w = 100$, $\ell = 150$, $A = 15000$

- The constraint is $3w+2\ell=600$. Substitute $\ell=(600-3w)/2$ into $A=w\ell$.
- Check feasibility, endpoints, and the sign or second derivative before declaring the optimum.

5. A poster has area 384 cm squared with 2 cm side margins and 3 cm top and bottom margins. Minimize total paper area.

Answer: *printed width* = 16, *printed height* = 24

- For printed width x , paper area is $(x+4)(384/x+6)$. Set its derivative to zero.
- Check feasibility, endpoints, and the sign or second derivative before declaring the optimum.

6. Find the closed cylinder of volume 250π with minimum surface area.

Answer: $r = 5$, $h = 10$

- Use $h=250/r$ to the power 2 in $S=2\pi r^2 + 2\pi rh$; solve $S'=0$.
- Check feasibility, endpoints, and the sign or second derivative before declaring the optimum.

7. Find the point on $y=x$ squared closest to $(0,3)$.

Answer: $(\pm\sqrt{5/2}, 5/2)$

- Minimize D to the power 2 $= x$ to the power 2 $+(x$ to the power 2 $-3)$ to the power 2. Critical points satisfy $2x(2x$ to the power 2 $-5)=0$; compare values.

- b. Check feasibility, endpoints, and the sign or second derivative before declaring the optimum.

8. Maximize the area of a rectangle inscribed under $y = \text{square root of } (25 - x \text{ to the power } 2)$.

Answer: $\text{width} = 5\sqrt{2}$, $\text{height} = 5/\sqrt{2}$, $A = 25$

- a. $A = 2x \text{ square root of } (25 - x \text{ to the power } 2)$; maximize $A^2 = 4x^2(25 - x^2)$.
b. Check feasibility, endpoints, and the sign or second derivative before declaring the optimum.

9. Cut 20 m of wire into a square and a circle to minimize total area.

Answer: $\text{square length} = \frac{80}{4+\pi}$, $\text{circle length} = \frac{20\pi}{4+\pi}$

- a. Let x go to the square: $A = x^2/16 + (20 - x)^2/(4\pi)$. Solve $A' = 0$.
b. Check feasibility, endpoints, and the sign or second derivative before declaring the optimum.

10. Demand is $p = 80 - 2q$. Maximize revenue.

Answer: $q = 20$, $p = 40$, $R = 800$

- a. $R = q(80 - 2q)$; $R' = 80 - 4q = 0$ and $R'' < 0$.
b. Check feasibility, endpoints, and the sign or second derivative before declaring the optimum.

11. Minimize average cost $C(q)/q$ when $C(q) = q^2 + 100q + 2500$.

Answer: $q = 50$

- a. Average cost is $q + 100 + 2500/q$. Set $1 - 2500/q^2 = 0$.
b. Check feasibility, endpoints, and the sign or second derivative before declaring the optimum.

12. A 10 ft ladder rests against a wall. Maximize the area of the right triangle it forms.

Answer: $x = y = 5\sqrt{2}$, $A = 25$

- a. With $x^2 + y^2 = 100$, maximize $A = xy/2$; symmetry or differentiation gives $x = y$.
b. Check feasibility, endpoints, and the sign or second derivative before declaring the optimum.

13. A cone has volume 72π . Minimize slant-independent material $S = \pi r^2 + \pi rh$.

Answer: $r = \sqrt[3]{72}$, $h = 3\sqrt[3]{72}$

- a. Use $h = 216/r^2$, then minimize $S = \pi r^2 + 216\pi/r$.
b. Check feasibility, endpoints, and the sign or second derivative before declaring the optimum.

14. A Norman window has perimeter 20. Maximize area.

Answer: $r = \frac{20}{4+\pi}$

- a. Use width $2r$ and rectangular height h . The perimeter gives $h = (20 - (2 + \pi)r)/2$; maximize $2rh + \pi r^2/2$.
b. Check feasibility, endpoints, and the sign or second derivative before declaring the optimum.

15. A boat 3 km offshore must reach a town 8 km downshore. Row at 3 km/h and walk at 5 km/h. Minimize time.

Answer: land about 5.75 km before town

- a. Let x be the downshore rowing distance. $T = \text{square root of } (x^2 + 9)/3 + (8 - x)/5$; solve $T' = 0$.
- b. Check feasibility, endpoints, and the sign or second derivative before declaring the optimum.

16. Maximize $f(x) = x(6 - x)$ on the closed interval $[1, 5]$.

Answer: $x = 3$, $f(3) = 9$

- a. Check the critical point $x = 3$ and both endpoints; the largest listed value is 9.
- b. Check feasibility, endpoints, and the sign or second derivative before declaring the optimum.